Homework

Question 22: *(From Hofmann, Ch. 3)* Let $f \in \mathbb{B}^+(Q)$ for some AFA's state set Q.

(a) Suppose $N \subseteq M \subseteq Q$ with $N \models f$. Show that $M \models f$ as well. Hint: use induction on the structure of positive Boolean formulae.

Proof. We will proceed by induction on the structure of positive Boolean formulae.

Base case: $f = q$ for some $q \in Q$. By the definition of satisfaction of Boolean formulae, $N \models q \iff q \in N$. Since it is given that $N \models q$, we have $q \in N$. Finally, since $N \subseteq M$, we can conclude that $q \in M$.

Inductive case: $f = g \vee h$ for some $g, h \in \mathbb{B}^+(Q)$. It is given that $N \models f = g \vee h$, and hence $N \models g$ or $N \models h$. In either case, a single application of the inductive hypothesis gives us satisfaction by M, and hence $M \models g \vee h$.

Inductive case: $f = g \wedge h$ for some $g, h \in \mathbb{B}^+(Q)$. We have as a given that $N \models f = g \wedge h$, and therefore $N \models g$ and $N \models h$. Applying the inductive hypothesis to each of these cases separately yields $M \models g$ and $M \models h$. Therefore, $M \models g \wedge h$. □

(b) Argue whether or not it is possible that $\varnothing \models f$ holds.

Proof. W.l.o.g., assume that f is an individual state. Then $\varnothing \models f \iff f \in \varnothing$. Clearly this statement is a contradiction, as $f \notin \varnothing$ for any f. Therefore, $\varnothing \not\models f$ for any f. □

(c) Argue whether or not it is possible that $Q \not\models f$ holds.

Proof. We have given that $f \in \mathbb{B}^+(Q)$. In other words, f is a positive Boolean formula over states of Q. Suppose $Q \not\models f$. Then by the definition of satisfaction, when all the states $q \in Q$ are set to *true* and all the states $q \notin Q$ are set to *false*, f evaluates to *false* under the usual rules of Boolean formula evaluation. But since f contains only states of Q and *positive* Boolean operators (*i.e.*, no negation), the truth of all states $q \in Q$ is enough to guarantee the truth of f. We can conclude that $Q \models f$ for all $f \in \mathbb{B}^+(Q)$. □

Question 23: Let $M \subseteq Q$ for some AFA's state set Q. We call M a *minimal model* of $f \in \mathbb{B}^+(Q)$ if $M \models f$ and $N \not\models f$ for all $N \subsetneq M$.

(a) Construct a positive Boolean formula that has more than one minimal model.

M is a minimal model of f if all states in M must be *true* for f to be *true*; in other words, if we make any state of M *false*, then f becomes *false* as well.

$M_1 = \{q_1\}$ and $M_2 = \{q_2\}$ are minimal models for the formula $f = q_1 \vee q_2$. It is clear from the definition of satisfaction that both M_1 and M_2 individually model f. Moreover, they are minimal models of f because their only proper subset is $\varnothing$, and we have already shown above in 22(b) that $\varnothing \not\models f$ for any f, and therefore in particular for $f = q_1 \vee q_2$.